

Additional Exercises 3

Power Series & Radius of Convergence

Mirrors: SIT3 | ★ Easy ★★ Medium ★★★ Hard

Exercise 1: Find the radius and domain of convergence

Use the ratio test: $R = \lim |a_n/a_{n+1}|$, or root test: $R = 1/\limsup \sqrt[n]{|a_n|}$. After finding R , check the endpoints $x = \pm R$ separately.

(a) ★ $\sum_{n=1}^{\infty} \frac{x^n}{3^n \cdot n}$

(b) ★ $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

(c) ★ $\sum_{n=1}^{\infty} (-1)^n \frac{x^n}{n}$

(d) ★ $\sum_{n=1}^{\infty} n! \cdot x^n$

(e) ★★ $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot x^{2n-1}}{5^n \sqrt{n+1}}$

(f) ★★ $\sum_{n=1}^{\infty} \frac{2^n \cdot x^n}{(2n+1)^2 \cdot \sqrt[4]{n}}$

(g) ★★ $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \cdot x^{2n-2}}{3^{n-1} \cdot n \sqrt{n}}$

(h) ★★ $\sum_{n=1}^{\infty} (-1)^n \ln(n) \cdot \frac{(2x+3)^n}{n \cdot (2n)!}$

(i) ★★★ $\sum_{n=1}^{\infty} \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)} \cdot x^n$

(j) ★★★ $\sum_{n=1}^{\infty} \frac{(2n)! \cdot x^n}{2^{2n} (n!)^2}$

(k) ★★★ $\sum_{n=1}^{\infty} \frac{(3n-1) \cdot (-1)^n \cdot x^n}{n^3}$

Exercise 2: Taylor series around $x = a$ — find and identify domain

- (a) ★ $f(x) = e^x$ around $a = 0$
- (b) ★ $f(x) = \sin x$ around $a = 0$
- (c) ★ $f(x) = \frac{1}{1-x}$ around $a = 0$
- (d) ★★ $f(x) = \frac{1}{x-5}$ around $a = 1$
- (e) ★★ $f(x) = \sin x$ around $a = \pi/2$
- (f) ★★ $f(x) = \ln x$ around $a = 1$
- (g) ★★★ $f(x) = \frac{1}{(x-5)^3}$ around $a = 1$ (*differentiate the series for $1/(x-5)$ term by term*)
- (h) ★★★ $f(x) = \frac{1}{(x-2)^2}$ around $a = -3$

Exercise 3: Expand into Maclaurin series

Use known series: e^u , $\sin u$, $\cos u$, $\ln(1+u)$, $\frac{1}{1-u}$ with substitution.

- (a) ★ $f(x) = e^{-x^2}$
- (b) ★ $f(x) = \sin(x^3)$
- (c) ★★ $f(x) = \frac{1}{x^2 + 4x + 5}$ (*complete the square*)
- (d) ★★ $f(x) = xe^x$
- (e) ★★ $f(x) = \ln(1+x^2)$
- (f) ★★★ $f(x) = \arctan\left(\frac{1+x^2}{1-x^2}\right)$
- (g) ★★★ $f(x) = \frac{2x^4 + 3x^3 - 15x^2 - 5x + 19}{x^2 + x - 6}$ (*polynomial division + partial fractions*)

Exercise 4: High-order derivatives via Taylor series

- (a) ★★ Compute $\frac{d^{17}}{dx^{17}} \left[\frac{1}{x-5} \right] \Big|_{x=1}$
- (b) ★★ Compute $\frac{d^{10}}{dx^{10}} \left[\frac{1}{1-x} \right] \Big|_{x=0}$

(c) ★★★ Given $f(x)$ analytic around a with radius $R_a > 1$. Find $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (1-a)^n$.

★ *Ratio test, known Maclaurin series* ★★ *Endpoint analysis, substitution into known series*
★★★ *Factorial products, term-by-term differentiation, abstract series.*